Junsei Ito

Curriculum Vitae

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早稲田大学 理工学術院 大学院先進理工学研究科 電気・情報生命専攻
システム制御研究室（和佐泰明研究室）
Department of Electrical Engineering and Bioscience,
Graduate School of Advanced Science and Engineering, Waseda University, Japan

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https://j-ito0625.github.io/
Google Scholar, researchmap, ORCID, GitHub, Qiita, LinkedIn

Bio: Junsei Ito is currently a second-year master's student in the Department of Electrical Engineering and Bioscience at Waseda University, Japan, under the supervision of Assoc. Prof. Yasuaki Wasa. He received the B.Eng. degree from Waseda University in 2025. He has been a JST ACT-X researcher since 2025. His research lies at the intersection of systems and control theory, information science, and physics, with interests in statistical learning theory, information geometry, and cybernetics. He is a member of the IEEE Control Systems Society (IEEE CSS), the Society of Instrument and Control Engineers (SICE), and the Japanese Society for Artificial Intelligence (JSAI).

Education and Research Positions

Research Positions

Oct. 2025 – present	JST ACT-X 研究者 Researcher, ACT-X, Japan Science and Technology Agency (JST)
Oct. 2025 – present	早稲田大学 理工学術院 和佐泰明研究室 リサーチアシスタント（RA） Research Assistant, Faculty of Science and Engineering, Waseda University (funded by JST ACT-X)

Education

Apr. 2025 – present	修士課程，早稲田大学大学院 先進理工学研究科 電気・情報生命専攻 **M.Eng.** course, Department of Electrical Engineering and Bioscience, Graduate School of Advanced Science and Engineering, Waseda University Advisor: Assoc. Prof. Yasuaki Wasa Master's Thesis: TBD
Apr. 2021 – Mar. 2025	学士（工学），早稲田大学 先進理工学部 電気・情報生命工学科 **B.Eng.**, Department of Electrical Engineering and Bioscience, School of Advanced Science and Engineering, Waseda University Advisor: Assoc. Prof. Yasuaki Wasa 論文題目：物理法則に基づく深層学習と自動微分を活用したデータ駆動型適応的 PID ゲイン最適化

Thesis Title: Data-Driven Adaptive PID Gain Optimization Using Physics-Based Deep Learning and Automatic Differentiation (in Japanese)

Apr. 2018 – Mar. 2021 宮城県仙台二華高等学校
Sendai Nika High School, Miyagi, Japan

Research Visits

Mar. 2024 大阪大学 御手洗光祐 准教授研究室 短期研究滞在
Short-term research stay, Osaka University (Host: Assoc. Prof. Kosuke Mitarai)
JAX を用いた超電導量子コンピュータのパルス最適化，自動微分による量子ゲート操作最適化

Honors and Awards

Mar. 2024 大阪大学 QIQB 量子ソフトウェア勉強会 優秀成績・特別選抜（93 名中 7 名）
Excellence Award and Special Selection (7/93), Osaka University QIQB Quantum Software Study Group
(Jun. 2023 – Mar. 2024)

Research Interests

Exploring the principles of intelligence and control under structural constraints in physical systems.

His research lies at the intersection of systems and control theory, information science, and physics. Physical systems carry structural constraints on observation, information, and actuation; these constraints are viewed not merely as technical obstacles but as fundamental conditions that shape what estimation, learning, and control can achieve. The research seeks to integrate the universal laws and interaction structures of physics with the data-driven learning and estimation capabilities of information science, within the framework of systems and control theory, and is also concerned with cybernetics and emergence, where adaptive and ordered behavior arises through local interactions and information network structures.

Keywords: Systems and Control Theory / Statistical Learning Theory / Information Geometry / Cybernetics

Research Supports and Grants

Grants

Oct. 2025 – Mar. 2028 JST ACT-X（研究代表者）
「部分観測下における Sim2Real 転移手法の開発」
Development of Sim2Real Transfer Methods under Partial Observation (Principal Investigator)
研究領域：［次世代 AI・数理情報］次世代 AI を築く数理・情報科学の革新
4,500,000 JPY + monthly salary（課題番号：JPMJAX25C3）
links: 早稲田大学プレスリリース, 日本の研究.com

Scholarships and Travel Grants

FY2026 早稲田大学 大学院生等海外派遣助成制度
Waseda University Overseas Research Travel Grant Program, 165,000 JPY

Publications

Preprints / Submitted Works

[P1] **Junsei Ito**, Yasuaki Wasa, "Data-Driven Adaptive PID Control Based on Physics-Informed Neural Networks," *IEEE Access*, under revision, 2025. [arXiv]

Peer-reviewed Conference Proceedings

[C2] **Junsei Ito**, Yasuaki Wasa, "Partial Observation Amplifies Model Mismatch in MAP Estimation via Information-Curvature Margins," *Proc. 65th IEEE Conference on Decision and Control*, accepted, HI, USA, Dec. 15–18, 2026. [arXiv]

[C1] **Junsei Ito**, Yasuaki Wasa, "Phase-Transition-Driven Distributed Coverage Control," *23rd IFAC World Congress*, accepted, Busan, Republic of Korea, Aug. 23–28, 2026. [simulation]

Domestic Conferences

[D5] **伊藤潤成**, 和佐泰明, "システム制御理論に基づく Physics-Informed Neural Networks の自律修復学習," 第 29 回 情報論的学習理論ワークショップ (IBIS2026), under preparation, 埼玉, 2026 年 10 月.

[D4] **伊藤潤成**, 和佐泰明, "フローマッチングによる移動ロボットの制御系設計," 2026 年度人工知能学会全国大会（第 40 回）論文集, 5I1-OS-3-02, 群馬, 2026 年 6 月. [DOI]

[D3] **伊藤潤成**, 和佐泰明, "相転移現象に基づく分散被覆制御," 第 13 回 計測自動制御学会 制御部門マルチシンポジウム予稿集, 1PS-43, 富山, 2026 年 3 月.

[D2] **伊藤潤成**, 和佐泰明, "群れの相転移現象に基づく動的環境適応のための自律分散マルチエージェント制御," 第 28 回 情報論的学習理論ワークショップ (IBIS2025), 2b-R-37, 沖縄, 2025 年 11 月.

[D1] **伊藤潤成**, 和佐泰明, "物理法則に基づく深層学習と自動微分を活用した適応的 PID ゲイン最適化," 第 12 回 計測自動制御学会 制御部門マルチシンポジウム予稿集, 2M7-4, 大阪, 2025 年 3 月.

Others

[O2] **伊藤潤成**, "システム制御理論に基づく Physics-Informed Neural Networks の自律修復学習," 計測自動制御学会 制御部門 自律型 Cyber-Physical Human Systems 調査研究会 CPHS 合宿 2026, under preparation, 群馬, 2026 年 11 月.

[O1] **伊藤潤成**, "物理法則に基づく深層学習と自動微分を活用したデータ駆動型適応的 PID ゲイン最適化," 早稲田大学 先進理工学部 電気・情報生命工学科 学士論文, 2025 年 3 月.

Teaching Activities

Teaching Assistant

2024, 2025　電気・情報生命工学実験 B (2024, 2025)
Electrical Engineering and Bioscience Laboratory B

2024　システム解析 (2024)
Systems Analysis

Professional Activities

Memberships

IEEE Control Systems Society
計測自動制御学会 (SICE)
人工知能学会 (JSAI)

Academic Committees

Jan. 2026 – present　計測自動制御学会 制御部門自律型 Cyber-Physical Human Systems 調査研究会，学生委員
Student Committee Member, SICE Control Division Research Committee on Cyber-Physical Human Systems for Autonomy

Industry Experience

Apr. 2026 – present　株式会社 Titan Intelligence, Researcher

Mar. 2024 – Oct. 2025　株式会社 QunaSys，量子計算ソフトウェア開発
Quantum computing software development, QunaSys Inc.